

Amplifier Classes



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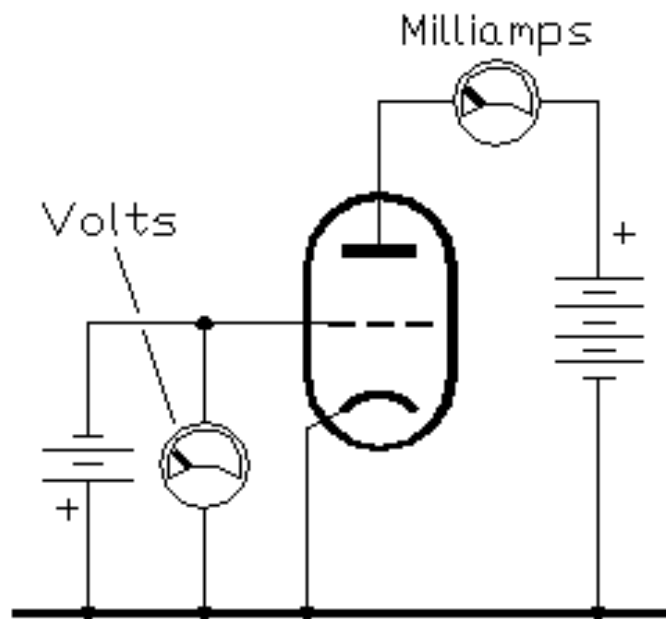
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You may hear of amplifiers being class A, AB1, AB2, B, C etc. What do all the characters and numbers really mean?

Let's forget the numbers for a moment - we will come back to them later. For now, we will look at the A/B/C issues, and in order to do so, we need to recap on how valves (or tubes) actually work.

Simple Valve Primer



If you know how valves work, then you may as well skip to Classes of Operation...

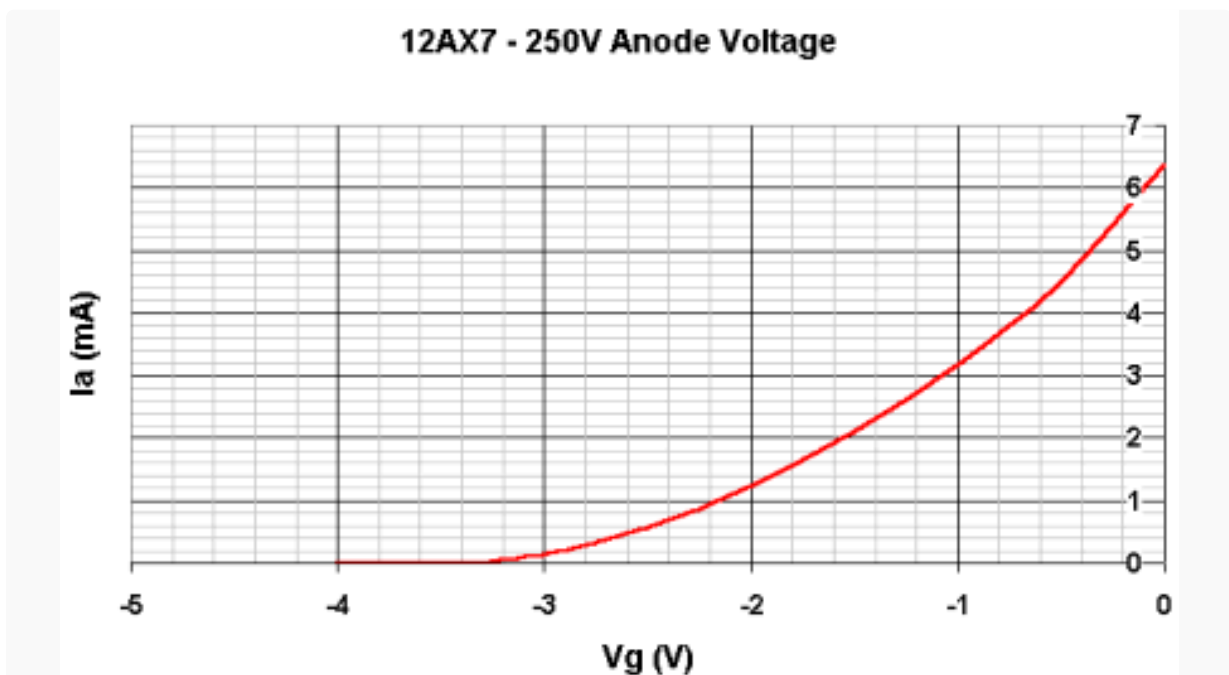
In the diagram on the right, we have connected a triode to a couple of batteries and a couple of meters.

Starting at the left, there is a battery connected between grid and ground - a voltmeter is there to measure how much voltage the battery is supplying to the grid. Note that the battery is "upside down", so the grid voltage will be negative compared to ground.

On the right, we have a larger battery with a milliammeter to measure how much current is being drawn by the anode (or plate) of the valve.

Typically, the grid might be negative by one or two volts for a preamp valve, and 40 volts or more for an output valve - this is known as the **bias setting** for the valve.

Amplification



By varying the voltage on the grid, we can vary the current going into the anode - this is how we achieve amplification.

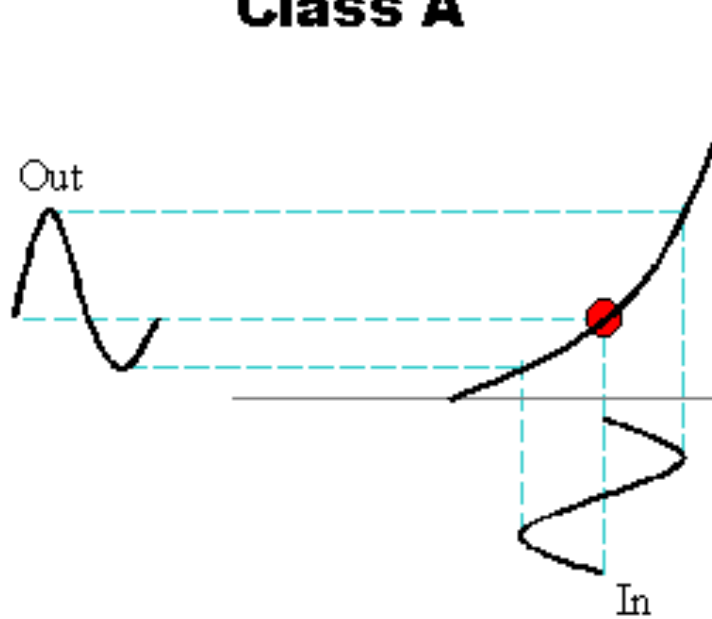
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Take a look at the chart on the right - this shows what happens to our valve as the grid voltage is varied. As the grid voltage is increased, the anode current increases.

The important point is that a valve converts a voltage input into a current output.

Whether it's a preamp valve, or an output valve, the principles remain the same. Now that we have recapped on how they work, we can look at the different classes of operation for output valves...

Class A



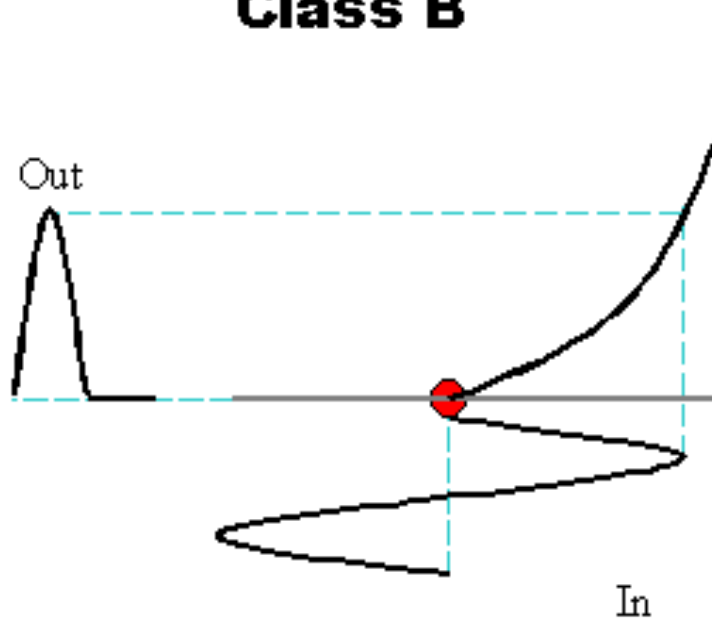
Have a look at the diagram on the right. The curve is the one taken from the previous section, and the bias has been set where the red spot is, roughly in the middle of the curve.

If we now superimpose a signal on the grid voltage, the anode current will vary up and down in sympathy with the grid voltage, hence the "Out" waveform.

As the title suggests, this is "Class A". The distinguishing feature of class A is that the valve is conducting current **at all times**. Note the the "Out" current never drops down to the zero line at any time.

Some output stages are class A (such as the Vox AC-30), and all preamp sections are class A. If you want to check out a circuit diagram with a class A output stage, you can [download](#) the circuit of my Blues-112 combo. Note that you will need Adobe Acrobat installed to make sense of it!

Class B

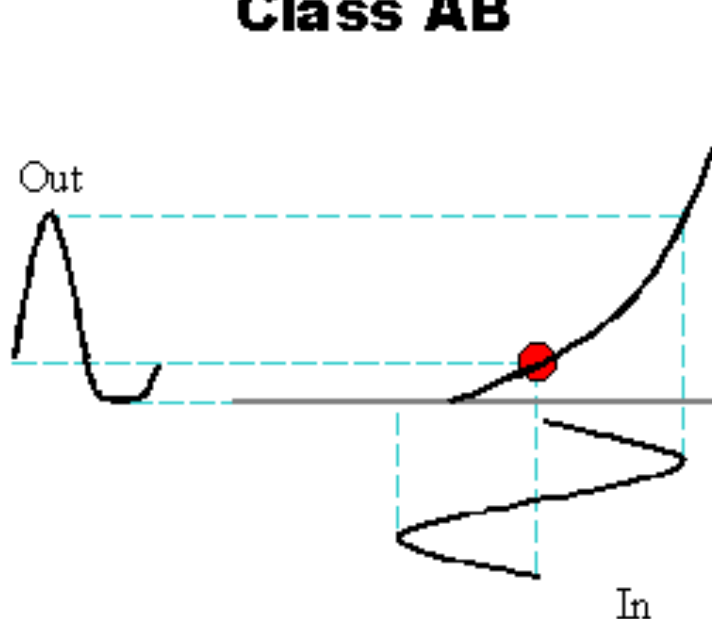


OK, that's class A out of the way. What about class B? In the diagram on the right, we have set the bias point to where the valve has almost stopped conducting.

Note that the input signal is a lot larger now in order to drive the valve hard enough. Also, the output current is only for half of the waveform.

To make any use of this, we have to have a **"push-pull"** output stage which employs two valves (or two banks of valves) so that each side amplifies each half of the waveform. While the first output valve provides the output current as shown on the right, the second valve fills in the gap which follows it.

Class AB



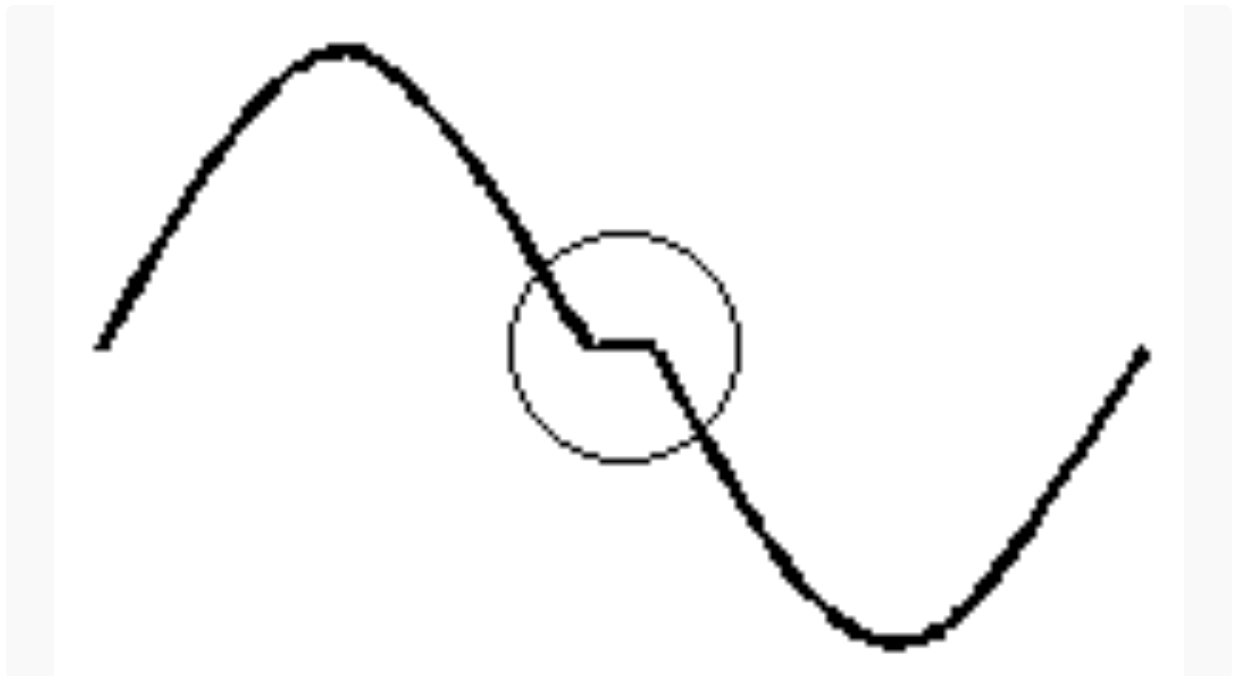
By now, you have probably guessed what class AB is - it's somewhere between class A and class B. Where exactly, is up to the imagination!

In our class AB diagram, a small amount of bias current is flowing through the valve. For the output valves in a typical class AB guitar amplifier, this would amount to around 30-40mA, with peaks of approximately 250-300mA.

In the push-pull output stage, there is a little overlap as each valve assists it's neighbour during a short transition, or crossover period.

Many larger guitar amplifiers are class AB, and we'll find out why a little later on.

Crossover Distortion



Class B suffers from a fundamental problem in that the push-pull amplifier does not, in practice, move smoothly from one half of the waveform to the other half.

When the current outputs from the two valves are added together by the output transformer, a kink can be seen on class B amplifiers (diagram on right). Class AB amplifiers can suffer from this also, if the bias current is too low.

In summary:

- Class B amplifiers *generally* introduce some crossover distortion
- Class AB amplifiers *may* introduce some crossover distortion
- Class A amplifiers introduce **no** crossover distortion, as both valves conduct simultaneously

The solution therefore, seems to be to use Class A for guitar amplifiers. Then we don't need to worry about crossover distortion at all. But...

Efficiency

Class AB and B exist for a reason, and the word above says it all. If we look back at the class A diagram, the output valve is drawing current all the time. To get the most power out of a class A amp, it is generally biased so that the bias current puts as much power through the valve as it can take.

This is not as efficient as the class B design which only draws current when it needs to, or the AB which is a hybrid somewhere between the two...

The lower efficiency of the class A means that more heat is generated for the same output power. Let's do a comparison of class A on a pair of EL34's and class B on the same pair of valves.

Manufacturers data for the EL34 shows a single ended class A design running the EL34 right on its anode dissipation limit of 25 watts. 11 watts of output is claimed at 10% THD. For a push-pull class A amplifier, the output figure would double due to the use of two valves, to 22 watts.

Further down the sheet, parameters for class B are listed. Claimed output is 100W at 5% THD - this is substantially higher than the 22 watts for the class A scenario!

Putting this into perspective, you would need **ten** EL34's to power a 100W amp in class A, and only two for class B. In the hybrid AB amplifiers, four EL34's are generally used for a 100W head such as a Marshall.

Other Classes

There are other classes such as class C, which shift the grid bias even more to the left. These classes are not useful for guitar amplification, and are best suited to Radio Frequency applications which use tuned circuits to remove distortion.

The Numbers

Oh yes, those numbers. There's A1 A2 AB1 AB2 etc., and these distinctions apply only to valve amplifiers.

The scheme is a simple one, with a "1" indicating that the valve does not draw any grid current, and "2" indicating that the valve output stage grid voltage is being pushed above the 0 volt mark and into a positive grid voltage, causing the grid to draw some current from the preceding driver stage.

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